MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN,
SILICON, POWER TYPES 2N6350, 2N6351, 2N6352, and 2N6353
JAN, JANTX, AND JANTXV

This specification is approved for use by US Army Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, Darlington, silicon, power transistors for use in high-speed power-switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. Types 2N6350 and 2N6351, see figure 1 (TO-33).
Types 2N6352 and 2N6353, see figure 3 (3 PIN TO-66).

1.3 Maximum ratings.

Type	P_T T_A = 25°C	P_T T_C = 100°C	V_{EB1}	V_{EB2}	V_{CBO1}	V_{CER}	I_C	I_C 1/	I_{B1}	T_{STG} and T_{op}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	V dc	A dc	A dc	A dc	°C	°C/W
2N6350	1.0 2/	5 3/	12	6	80	80	5	10	0.5	-65	20
2N6351	1.0 2/	5 3/	12	6	150	150	5	10	0.5	to	20
2N6352	2.0 4/	25 5/	12	6	80	80	5	10	0.5		4.0
2N6353	2.0 4/	25 5/	12	6	150	150	5	10	0.5	+200	4.0

1/ Applies for $t_p \leq 10$ ms, duty cycle $\leq 50\%$.

2/ Derate linearly 5.72 mW/°C for $T_A > 25$°C.

3/ Derate linearly 50 mW/°C for $T_C > 100$°C.

4/ Derate linearly 11.4 mW/°C for $T_A > 25$°C.

5/ Derate linearly 250 mW/°C for $T_C > 100$°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703, using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_A = 25°C.

	h_{FE}		$|h_{fe}|$	$V_{BE1(on)}$	$V_{CE(sat)}$		Pulse response	
							t_{on}	t_{off}
	V_{CE} = 5 V dc I_C = 5 A dc		V_{CE} = 10 V dc I_C = 1 A dc f = 10 MHz	V_{CE} = 5 V dc I_C = 5 A dc	I_C = 5 A dc I_{B1} = 5 mA dc	I_C = 5 A dc I_{B1} = 10 mA dc	I_C = 5 A dc	I_C = 5 A dc
Limit	2N6350 2N6352	2N6351 2N6353			2N6350 2N6352	2N6351 2N6353		
				V dc	V dc	V dc	µs	µs
Min	2,000	1,000	5	---	---	---	---	---
Max	10,000	10,000	25	2.5	1.5	2.5	0.5	1.2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions as shown on figures 1 and 3.

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3.3.1 Lead material and finish. Lead material, for types 2N6350 and 2N6351, shall be Kovar or Alloy 52. Lead finish shall be gold-plated or tin-coated. Where a choice of lead material or finish is desired, it shall be specified in the contract or order (see 6.2).

3.3.2 Terminal-lead length (types 2N6350 and 2N6351). Terminal-lead length(s) other than that specified in figure 1 may be furnished when so stipulated in the contract or order where the devices covered herein are required directly for particular equipment-circuit installation or for automatic-assembly-technique programs. Where other lead lengths are required and provided, it shall not be construed as adversely affecting the qualified product status of the device, or applicable JAN marking (see 6.2).

3.4 Marking. Device marking shall be in accordance with MIL-S-19500. For types 2N6350 and 2N6351, the following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

a. Country of origin

b. Manufacturer's identification

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

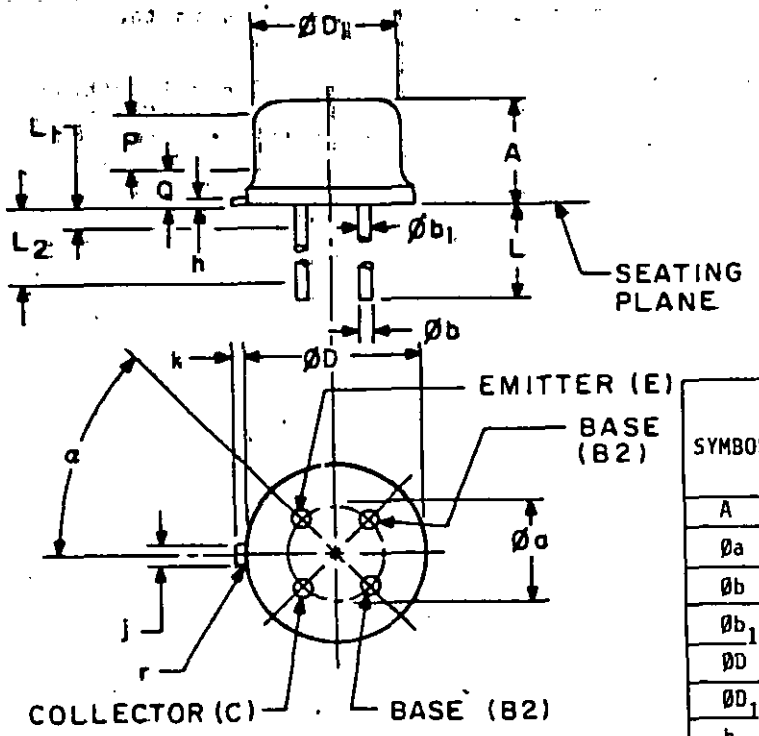
4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

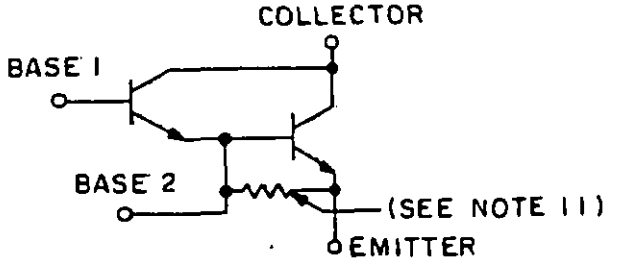
Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels
7	See 4.5.2
9	Not applicable
11	I_{CEX1} and h_{FE1}
12	See 4.3.1
13	Subgroup 2 of table I herein: ΔI_{CEX1} = ±100 percent or 20 nA, whichever is greater. Δh_{FE1} = ±25%.

4.3.1 Power burn-in test conditions. Power burn-in conditions (at T_A = 25°C) are as follows:

2N6350	V_{CE} = 60 V dc	P_T = 1.0 W
2N6351	V_{CE} = 100 V dc	P_T = 1.0 W
2N6352	V_{CE} = 60 V dc	P_T = 2.0 W
2N6353	V_{CE} = 100 V dc	P_T = 2.0 W

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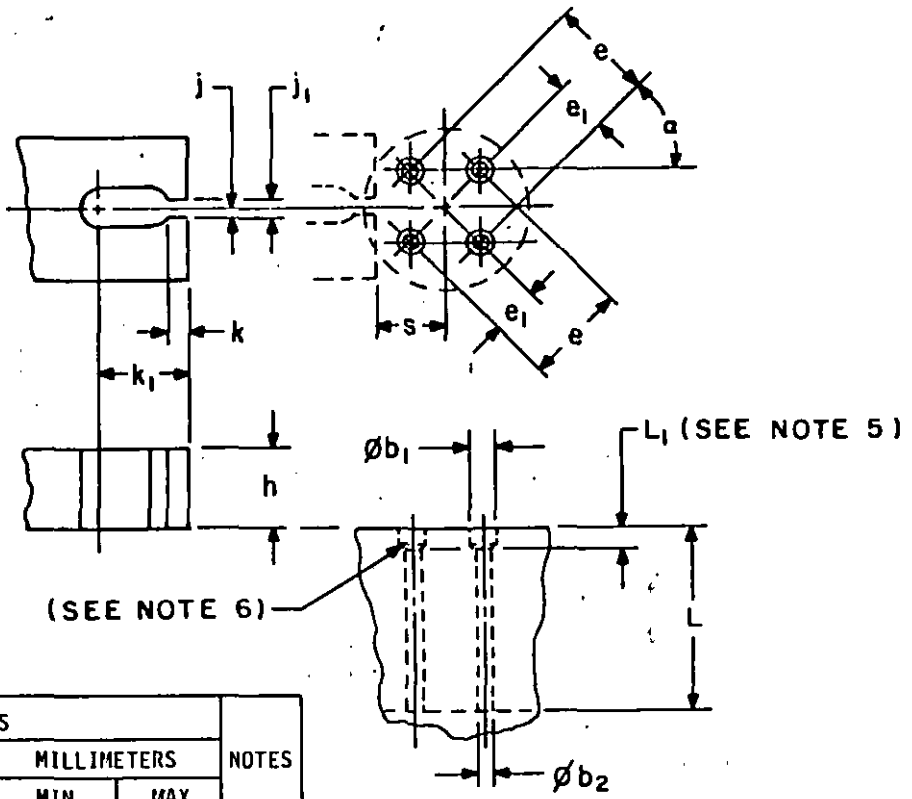
SCHEMATIC CIRCUIT

SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.184	.260	4.67	6.60	
Øa	.200 TP		5.080 TP		6
Øb	.016	.021	.407	.533	7,8
$Øb_1$	.016	.019	.407	.482	7,8
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
h	.009	.041	.23	I.04	
j	.028	.034	.712	.863	2
k	.029	.045	.74	1.14	3
L	1.500	1.750	38.10	44.45	7,8
L_1		.050		1.27	7,8
L_2	.250		6.35		7,8
P	.100		2.54		5
Q		.050		1.27	4
r		.010		.254	10
a	45° TP		45° TP		6

NOTES:
1. Metric equivalents are given for general information only.
2. Beyond r(radius) maximum, j shall be held for a minimum length of .011 (0.28 mm).
3. k measured from maximum ØD.
4. Outline in this zone is not controlled.
5. $ØD_1$ shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001 -.000 (1.37 +.03 -.00 mm) below seating plane shall be within .007 (0.18 mm) radius of True Position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 3.
7. $Øb_1$ applies between L_1 and L_2. Øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. The collector shall be electrically connected to the case.
10. r(radius) applies to both inside corners of tab.
11. Internal resistance (typically 750 ohms).

FIGURE 1. Physical dimensions of transistor types 2N6350 and 2N6351 (TO-33).

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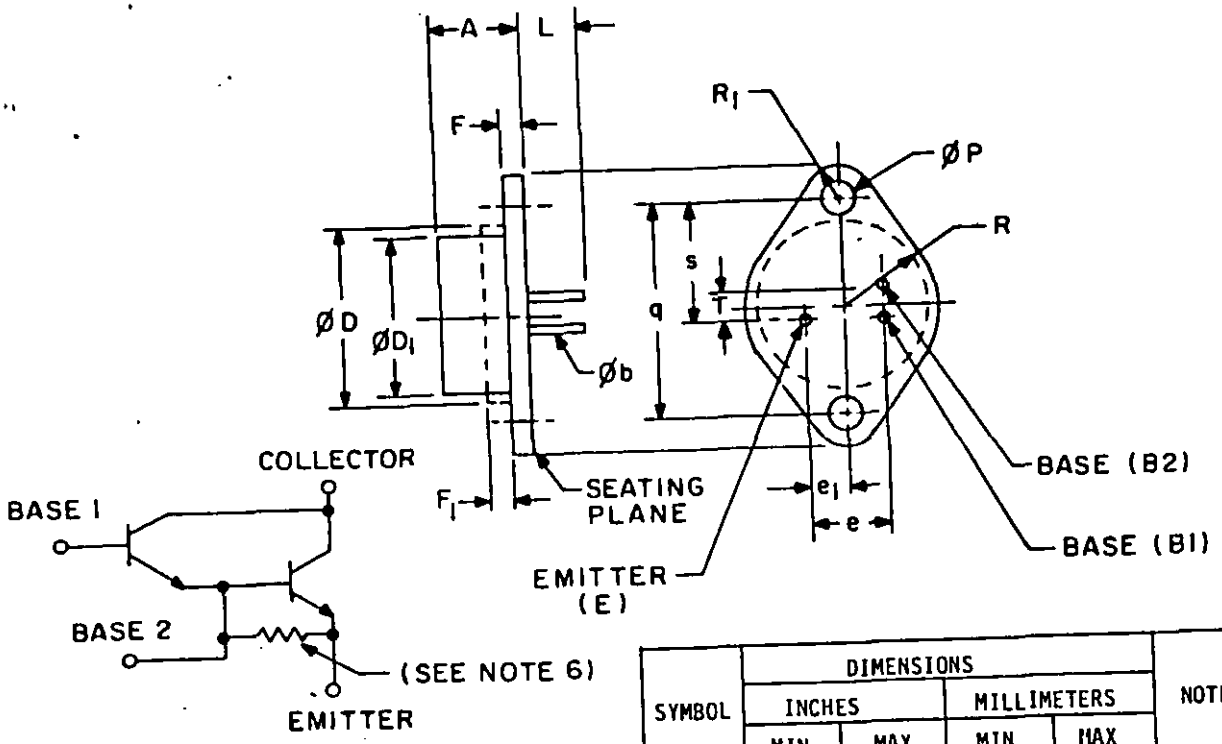
SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
$øb_1$	.0595	.0605	1.511	1.537	
$øb_2$	.0325	.0335	.8255	.8509	
e	.1995	.2005	5.067	5.093	
e_1	.0995	.1005	2.527	2.552	
h	.150 NOM		3.81 NOM		
j	.0175	.0180	.4445	.4572	
j_1	.0350	.0355	.8890	.9017	
k	.009	.011	.23	.28	
k_1	.125 NOM		3.18 NOM		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. The following gauging procedure shall be used:

The device being measured shall be inserted until its seating plane is .125 (3.18 mm) ±.010 (0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the devices cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.

FIGURE 2. Gauge for lead and tab location for transistor types 2N6350 and 2N6351.

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SCHEMATIC CIRCUIT

SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.250	.340	6.35	8.64	
Øb	.028	.034	.711	.863	4
ØD		.620		15.75	
$ØD_1$	.470	.500	11.94	12.70	
e	.190	.210	4.83	5.33	
e_1	.093	.107	2.36	2.72	
F	.050	.075	1.27	1.91	2
F_1		.050		1.27	1
L	.360	.500	9.14	12.70	4
Øp	.142	.152	3.61	3.86	
q	.958	.962	24.33	24.43	
R		.350		8.89	
R_1	.115	.145	3.92	3.68	
S	.570	.590	14.48	14.99	
T	.190	.210	4.83	5.33	

NOTES:
1. The outline contour is optional within zone defined by ØD and F_1.
2. Dimension does not include sealing flanges.
3. Metric equivalents are given for general information only.
4. Both terminals.
5. The collector shall be electrically connected to the case.
6. Internal resistance (typically 750 ohms).

FIGURE 3. Physical dimensions of transistor types 2N6352 and 2N6353 (TO-66).

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Hermetic seal (fine-leak) test. All devices shall be fine-leak tested in accordance with MIL-STD-750, method 1071, test condition G or H; except the leak-rate rejection criterion shall be 5×10^{-7} (for types 2N6352 and 2N6353), and 1×10^{-7} (for types 2N6350 and 2N6351) cubic centimeters of helium per second when measured at a differential pressure of one atmosphere.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents may specify the following:

a. Lead finish if other than specified (see 3.3.1)

b. Selectivity of lead material (see 3.3.1)

c. Terminal-lead length (see 3.3.2)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - ER

Review activity:
DLA - ES

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5961-A834)

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TABLE I. Group A inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD JAN JANTX JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 1			5				
Visual and mechanical inspection	2071						
Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias cond. B; I_C = 25 mA dc pulsed (see 4.5.1) R_{B1E} = 2.2 kΩ R_{B2E} = 100Ω		$V_{(BR)CER}$			
2N6350, 2N6352					80	---	V dc
2N6351, 2N6353					150	---	V dc
Breakdown voltage, emitter to base 1	3026	Bias cond. D; I_E = 12 mA dc Base 2 open		$V_{(BR)EBO1}$	12	---	V dc
Breakdown voltage, emitter to base 2	3026	Bias cond. D; I_E = 12 mA dc Base 1 open		$V_{(BR)EBO2}$	6	---	V dc
Collector to emitter cutoff current	3041	Bias cond. A; V_{EB1} = 2 V dc R_{B2E} = 100Ω		I_{CEX1}			
2N6350, 2N6352		V_{CE} = 80 V dc			---	1.0	μA dc
2N6351, 2N6353		V_{CE} = 150 V dc			---	1.0	μA dc
Collector to emitter voltage (saturated)	3071	I_C = 5.0 A dc R_{B2E} = 100Ω		$V_{CE(sat)}$			
2N6350, 2N6352		I_{B1} = 5 mA dc pulsed (see 4.5.1)			---	1.5	V dc
2N6351, 2N6353		I_{B1} = 10 mA dc pulsed (see 4.5.1)			---	2.5	V dc
Base to emitter voltage	3066	I_C = 5.0 A dc pulsed (see 4.5.1); V_{CE} = 5 V dc; R_{B2E} = 100Ω		$V_{BE1(on)}$	---	2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1.0 A dc pulsed (see 4.5.1); R_{B2E} = 1 kΩ		h_{FE1}			
2N6350, 2N6352					2,000	---	---
2N6351, 2N6353					1,000	---	---

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions	JAN JANTX JANTXV		Min	Max	
Subgroup 2 - Continued			5				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5.0 A dc pulsed (see 4.5.1); R_{B2E} = 100Ω		h_{FE2}			
2N6350, 2N6352					2,000	10,000	---
2N6351, 2N6353					1,000	10,000	---
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 A dc pulsed (see 4.5.1); R_{B2E} = 100Ω		h_{FE3}			
2N6350, 2N6352					400	---	---
2N6351, 2N6353					200	---	---
Subgroup 3			5				
High temperature operation:		T_A = +150°C					
Collector to emitter cutoff current	3041	Bias cond. A, V_{EB1} = 2 V dc; R_{B2E} = 100Ω		I_{CEX2}			
2N6350, 2N6352		V_{CE} = 80 V dc			---	1.0	mA dc
2N6351, 2N6353		V_{CE} = 150 V dc			---	1.0	mA dc
Low temperature operation:		T_A = 65°C					
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5.0 A dc pulsed (see 4.5.1) R_{B2E} = 100Ω		h_{FE4}			
2N6350, 2N6352					400	---	---
2N6351, 2N6353					200	---	---
Subgroup 4			5				
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1.0 A dc f = 10 MHz; R_{B2E} = 100Ω		$\lvert h_{fe} \rvert$	5	25	---
Open circuit output capacitance	3236	V_{CB1} = 10 V dc; 100 kHz ≤ f ≤ 1 MHz Base 2 open		C_{obo1}	---	120	pF

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD JAN JANTX JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 4 - Continued			5				
Pulse response:							
Turn-on time	3251	Test cond. A; V_{CC} = 30 V dc; I_C = 5.0 A dc;		t_{on}			
2N6350, 2N6352		See figure 4			---	0.5	µs
2N6351, 2N6353		See figure 5			---	0.5	µs
Turn-off time	3251	Test cond. A; V_{CC} = 30 V dc; I_C = 5.0 A dc;		t_{off}			
2N6350, 2N6352		See figure 4			---	1.2	µs
2N6351, 2N6353		See figure 5			---	1.2	µs
Subgroup 5			10				
Safe operating area (continuous dc) (for types 2N6350 and 2N6351 only)	3051	T_C = +100°C; $t \geq 1$ s, 1 cycle; $t_r + t_f$ = 10 µs (see figure 6) R_{B2E} = 100Ω					
Test 1 2N6350, 2N6351		V_{CE} = 1.5 V dc; I_C = 3.3 A dc		---	---	---	---
Test 2 2N6350, 2N6351		V_{CE} = 30 V dc; I_C = 167 mA dc		---	---	---	---
Test 3 2N6350		V_{CE} = 80 V dc; I_C = 35 mA dc		---	---	---	---
Test 4 2N6351		V_{CE} = 150 V dc; I_C = 13 mA dc		---	---	---	---
Safe operating area (continuous dc) (for types 2N6352 and 2N6353 only)	3051	T_C = +100°C; $t \geq 1$ s 1 cycle; $t_r + t_f$ = 10 µs (see figure 7) R_{B2E} = 100Ω					
Test 1 2N6352, 2N6353		V_{CE} = 5.0 V dc; I_C = 5 A dc		---	---	---	---
Test 2 2N6352, 2N6353		V_{CE} = 10 V dc; I_C = 2.5 A dc		---	---	---	---

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions	JAN JANTX JANTXV		Min	Max	
Subgroup 5 - Continued			10				
Test 3 2N6352		V_{CE} = 80 V dc; I_C = 95 mA dc		---	---	---	---
Test 4 2N6353		V_{CE} = 150 V dc; I_C = 35 mA dc		---	---	---	---
Electrical measurements		See table IV, steps 1, 2, and 3					
Safe operating area (clamped switching)	3053	Load cond. B (inductive load) (see figure 9); T_A = + 100°C $t_r + t_f \leq 1.0$ us Duty cycle < 2%; t_p 8 ms; (vary to obtain I_C) R_S = 0.10Ω R_{BB1} = 200Ω V_{BB1} = 10 V dc R_{BB2} = 2.2 kΩ V_{BB2} = -4 V dc; I_C = 5 A dc; V_{CC} = 25 V dc $R_L \leq 2.5Ω$ L = 40 mH (Triad-C- 48U, or equivalent) R_{B2E} = 100Ω					
2N6350, 2N6352		Clamp voltage = 50 +0, -5 V dc					
2N6351, 2N6353		Clamp voltage = 90 +0, -5 V dc					
Electrical measurements		See table IV, steps 1, 2, and 3					
Subgroup 6							
Not applicable							
Subgroup 7							
Not applicable							

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TABLE II. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			10
Thermal shock (temperature cycling)	1051	Test condition C-1; time at temperature extremes = 15 minutes minimum	
Hermetic seal	1071		
a. Fine leak			
2N6350, 2N6351		Leak rate = 1 x 10^{-7} atm cm^3/s	
2N6352, 2N6353		Leak rate = 5 x 10^{-7} atm cm^3/s	
b. Gross leak			
Electrical measurements		See table IV, steps 1, 2, and 3	
Subgroup 3			5
Steady-state operation life	1026		
2N6350, 2N6351		T_A = +25°C; P_T = 1.0 W; V_{CE} = 40 V dc	
2N6352, 2N6353		T_C = +100°C; P_T = 25 W; V_{CE} = 10 V dc	
Electrical measurements		See table IV, steps 1, 2, and 4	
Subgroup 4			
Decap internal visual (design verification)	2075		1 device/0 failure for each lot
Bond strength	2037		20 (c = 0)
Subgroup 5			
Thermal resistance (junction to case)	3151	Base 1 shorted to base 2 (see 1.3)	15

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TABLE II. Group B inspection for JAN, JANTX, and JANTXV devices - Continued.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 6			7
High temperature life (nonoperating)	1032	T_{STG} = +200°C	
Electrical measurements		See table IV, steps 1, 2, and 3	
Subgroup 7			10
Safe operating area (unclamped switching)	3053	Load cond. C (inductive load) (see figure 8); T_A = +25°C, duty cycle ≤ 2% R_S = 0.10Ω; t_p 80 µs (vary to obtain I_C) $t_r + t_f \leq 15$ ns, R_{BB1} = 200Ω V_{BB1} = 10 V dc; R_{BB2} = 2.2 kΩ V_{BB2} = -4 V dc; I_C = 5 A dc; V_{CC} = 25 V dc; R_L = .07Ω L = 0.4 mH (48 turns 1 row of 20 ASG wire on Arnold Engineering Co. Core A-930157-2, or equivalent) R_{B2E} = 100Ω	
Electrical measurements		See table IV, steps 1, 2, and 3.	

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TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Physical dimensions	2066	See figures 1 and 2	
Subgroup 2			10
Thermal shock (glass strain)	1056	Test cond. B	
Terminal strength:	2036		
a. Lead fatigue (for types 2N6350 and 2N6351 only)		Test cond. E	
b. Tension (for types 2N6352 and 2N6353 only)		Test cond. A; weight = 3 lbs; time = 15s	
Hermetic seal	1071		
a. Fine leak			
2N6350, 2N6351		Leak rate = 1 x 10^{-7} atm cc/s	
2N6352, 2N6353		Leak rate = 5 x 10^{-7} atm cc/s	
b. Gross leak			
Moisture resistance	1021	Omit initial condition for 2N6352, 2N6353 only	
Visual and mechanical inspection	2071		
Electrical measurements		See table IV, steps 1, 2, and 3	
Subgroup 3			10
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1, 2, and 3	
Subgroup 4			15
Salt atmosphere (corrosion)	1041		

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TABLE III. Group C inspection (all quality levels) - Continued.

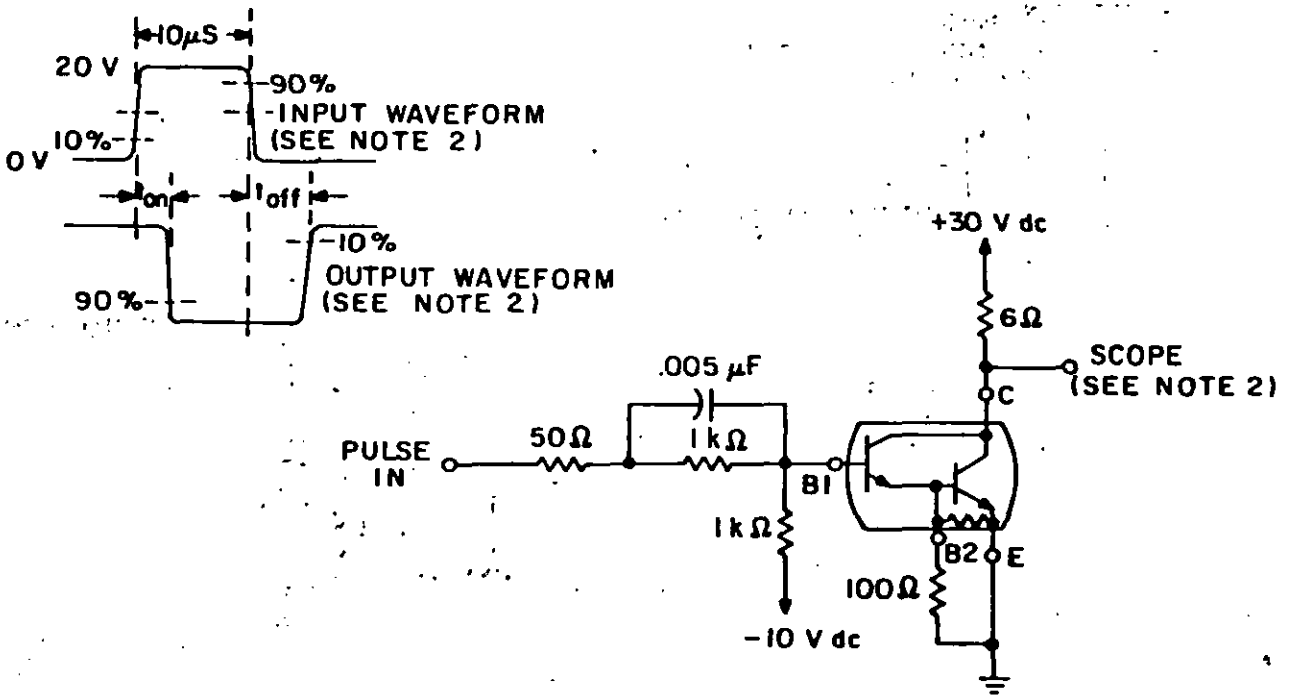
Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 5			
Not applicable			
Subgroup 6			λ = 10
Steady-state operation life	1026		
2N6350, 2N6351		T_A = +25°C; P_T = 1.0 W; V_{CE} = 40 V dc	
2N6352, 2N6353		T_C = +100°C; P_T = 25 W; V_{CE} = 10 V dc	
Electrical measurements		See table IV, steps 1, 2, and 3	

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TABLE IV. Group B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to emitter cutoff current	3041	Bias cond. A; V_{EB1} = 2 V dc; R_{B2E} = 100Ω	I_{CEX1}			
	2N6350, 2N6352		V_{CE} = 80 V dc;		---	1.0	µA dc
	2N6351, 2N6353		V_{CE} = 150 V dc;		---	1.0	µA dc
2	Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = 25 mA dc; pulsed (see 4.5.1); R_{B1E} = 2.2 kΩ R_{B2E} = 100Ω	$V_{(BR)CER}$			
	2N6350, 2N6352				80	---	V dc
	2N6351, 2N6353				150	---	V dc
3	Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5.0 A dc pulsed (see 4.5.1); R_{B2E} = 100Ω	h_{FE2}			
	2N6350, 2N6352				2,000	10,000	---
	2N6351, 2N6353				1,000	10,000	---
4	Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5.0 A dc pulsed (see 4.5.1); R_{B2E} = 100Ω	Δh_{FE2}	---	± 25% change from initial group A reading	

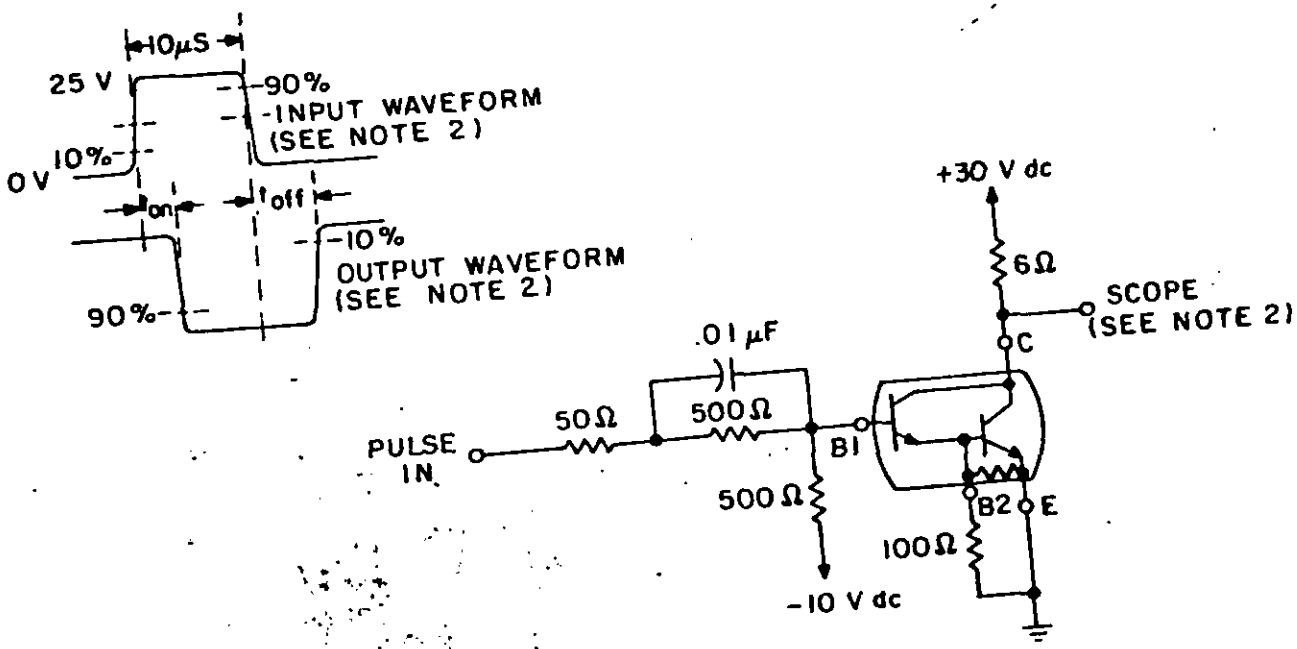
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NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{out} = 50\Omega$, PW = 10 μs, duty cycle ≤ 2%.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{in} \geq 10$ MΩ, $C_{in} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The DC power supplies may require additional by-passing in order to minimize ringing.

FIGURE 4. Pulse response test circuit for types 2N6350 and 2N6352.

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NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{out} = 50\Omega$, PW = 10 µs, duty cycle $\leq 2\%$.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{in} \geq 10$ MΩ, $C_{in} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The DC power supplies may require additional by-passing in order to minimize ringing.

FIGURE 5. Pulse response test circuit for types 2N6351 and 2N6353.

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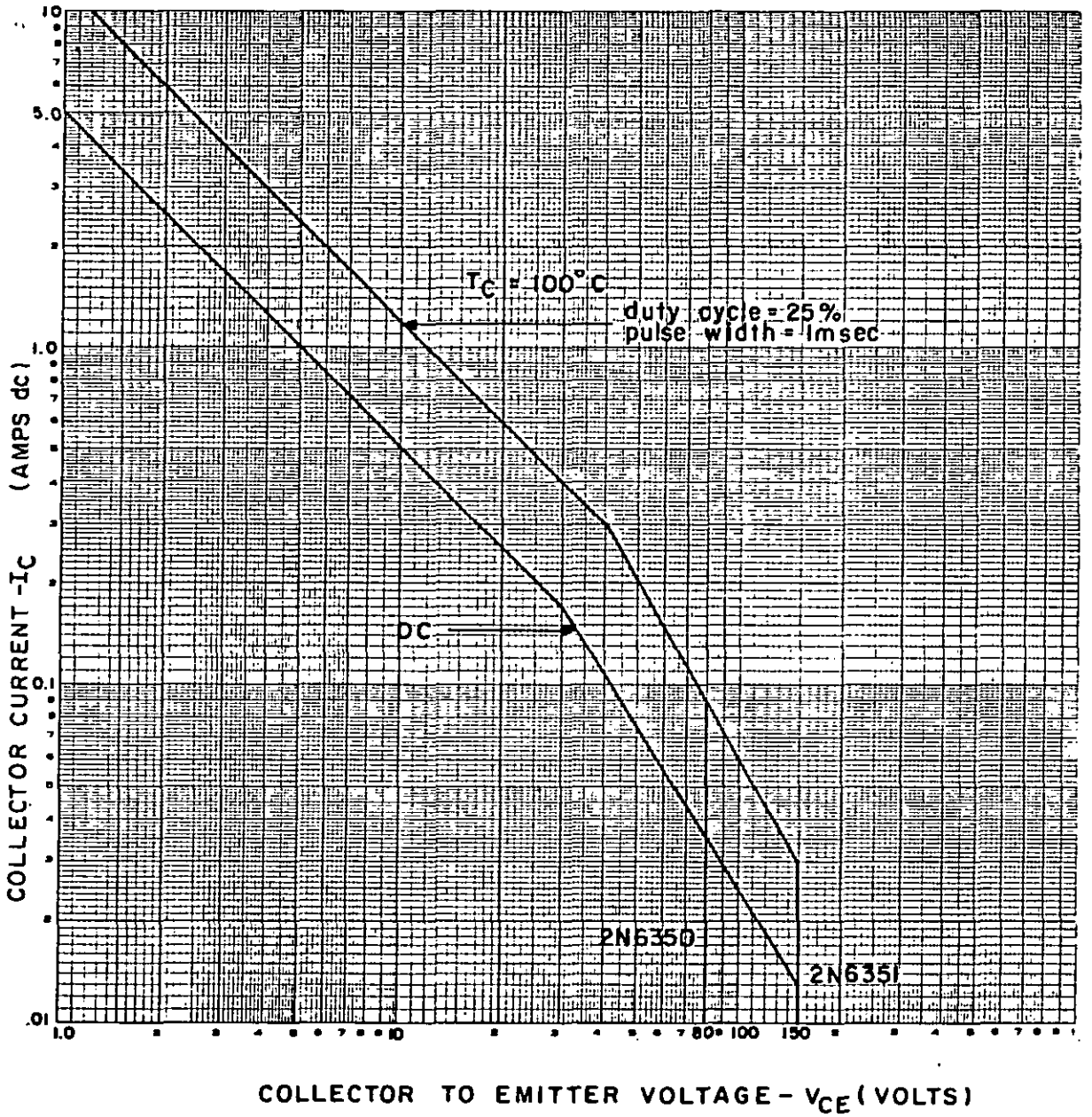


FIGURE 6. Maximum safe operating area graph for types 2N6350 and 2N6351.

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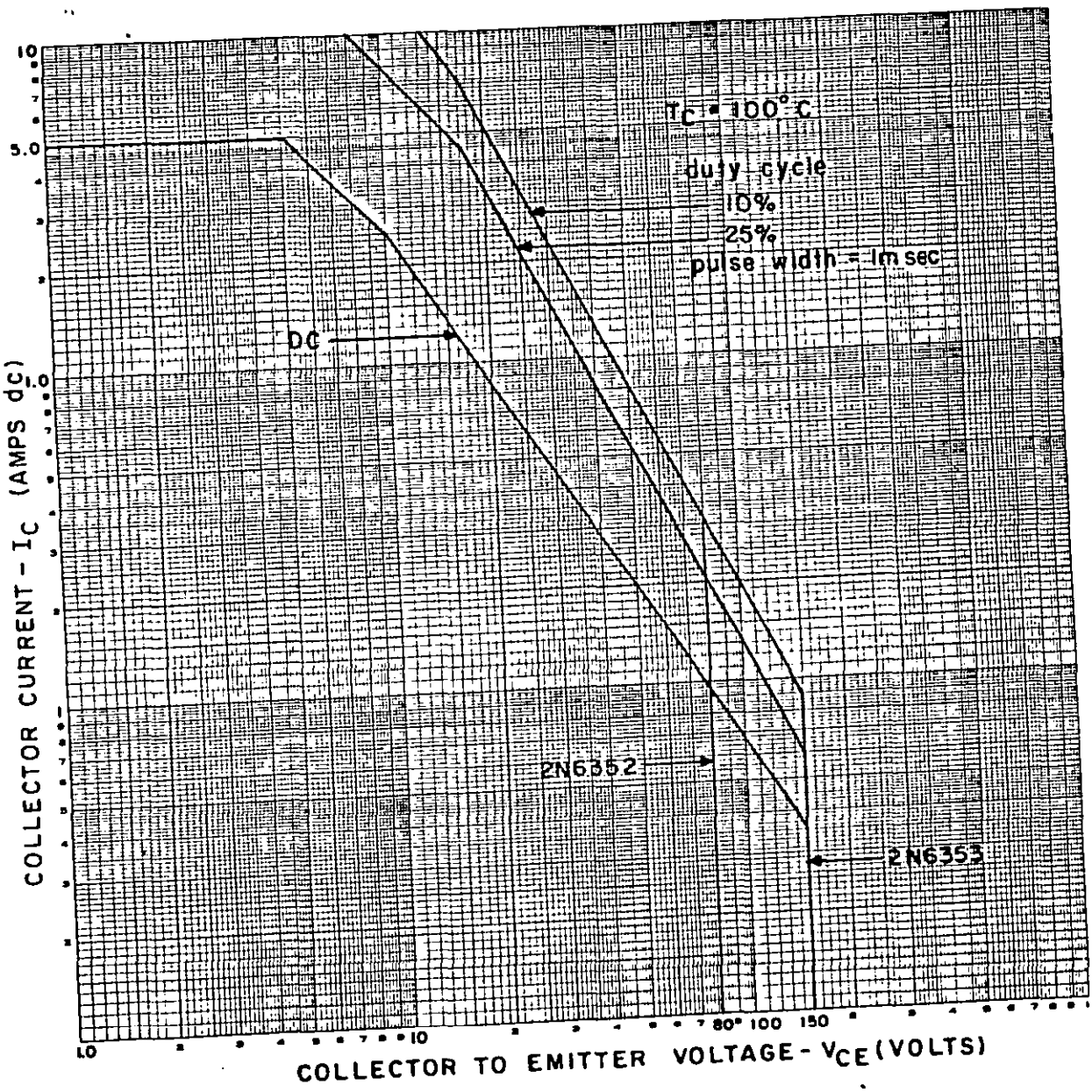


FIGURE 7. Maximum safe operating area graph for types 2N6352 and 2N6353.

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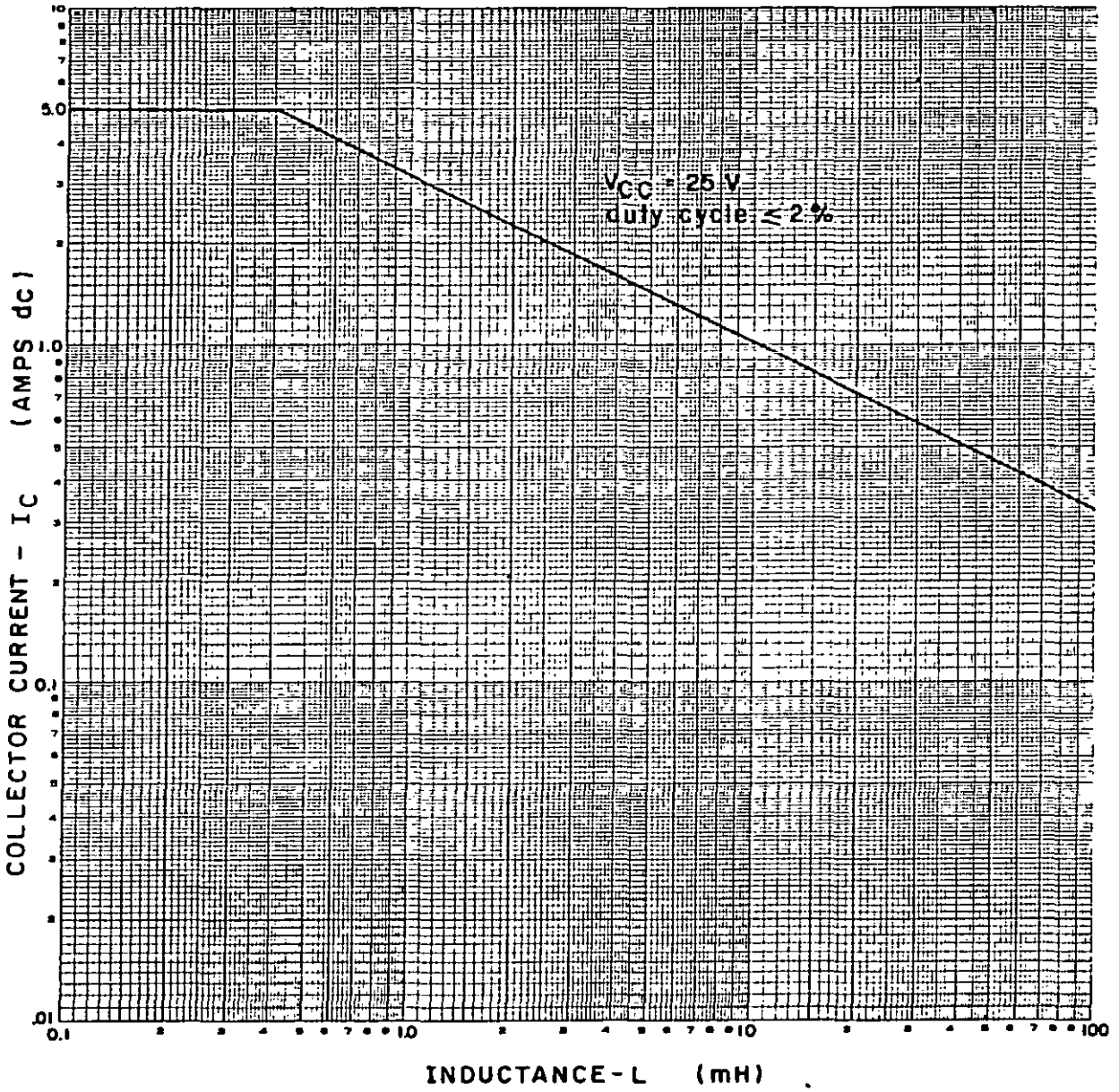


FIGURE 8. Safe operating area for switching between saturation and cutoff - (unclamped inductive load).

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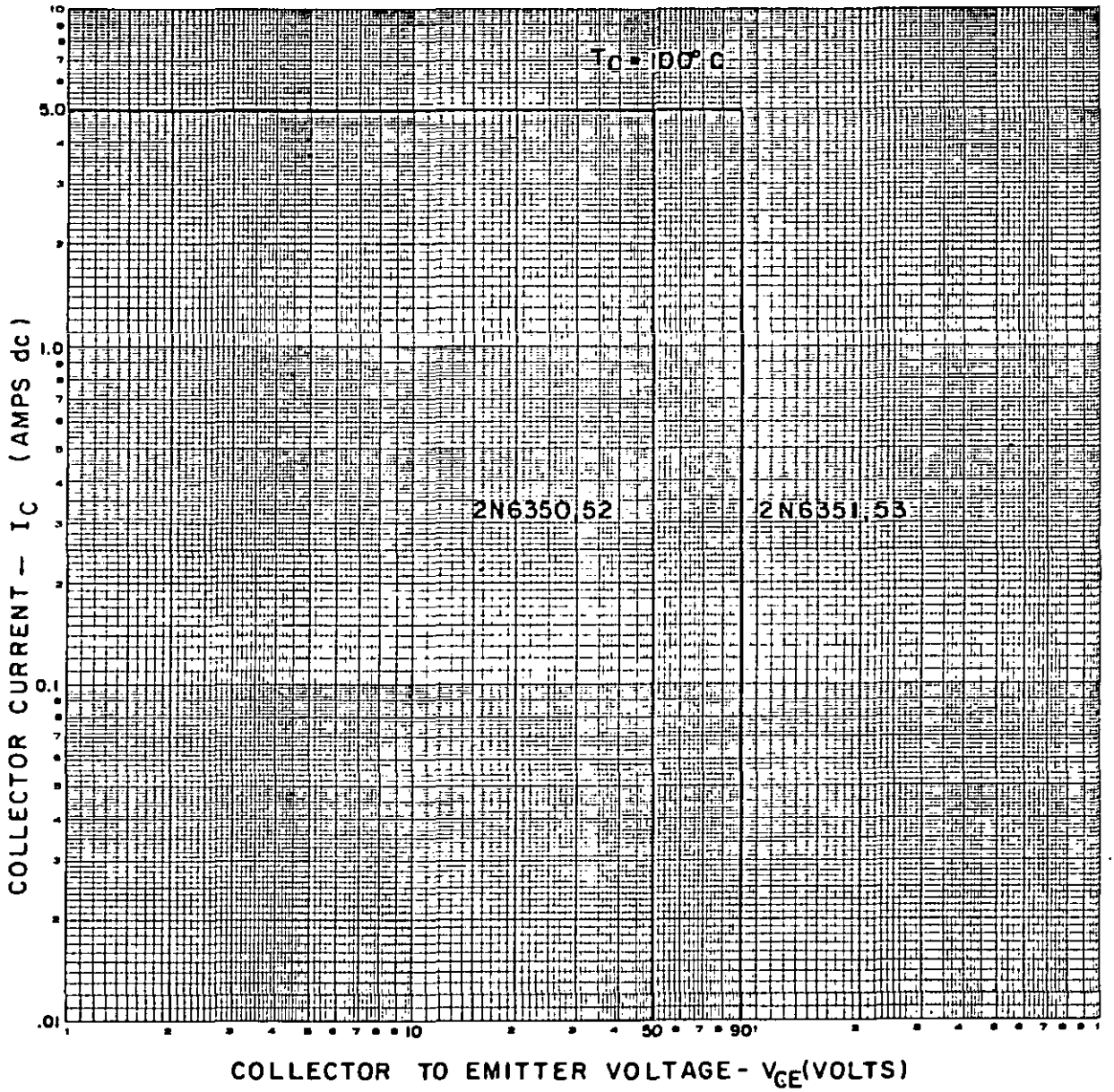


FIGURE 9. Safe operating area for switching between saturation and cutoff - (clamped inductive load).

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